100 Python Problems

CampusX • Practice Question Booklet

Questions only — no solutions included

Includes all 78 Python problem files found in the uploaded repository folder.

1. Largest of 3 nums

User will input (3 ages). Find the oldest one.

2. C to f temp

Write a program that converts Celsius values to Fahrenheit.

3. Swap vars w third

The user will input two numbers. Write a program to swap the numbers.

4. Sum of digits

Write a program that will give you the sum of three digits.

5. Reverse digits

Write a program that reverses a four-digit number. Additionally, ensure that it checks whether the reversal is true.

6. Even or odd

Write a program that will determine whether the number entered by the user is even or odd.

7. Leap year

Write a program that will determine whether the given year is a leap year or not.

8. Euclidean dist

Write a program to calculate the Euclidean distance between two coordinates.

9. 3 angles triangle

Write a program that takes user input for three angles and determines whether they can form a triangle or not.

10. Profit loss calc

Write a program that takes user input for cost price and selling price, and determines whether it's a loss or a profit.

11. Simple interest

Write a program to find the simple interest when the values of principal, rate of interest, and time period are given.

12. Vol of cylinder

Write a program to find the volume of the cylinder. Also, determine the cost when the price of 1 liter of milk is 40 Rs.

13. Div by 3 & 6

Write a program that will determine whether the given number is divisible by 3 and 6.

14. Weather type

Write a program that will determine whether the value of temperature and humidity is provided by the user. | TEMPERATURE(C) | HUMIDITY(%) | WEATHER | |----------------|-------------|----------------| | >= 30 | >= 90 | Hot and Humid | | >= 30 | < 90 | Hot | | < 30 | >= 90 | Cold and Humid | | < 30 | < 90 | Cold |

15. Sum of squares

Write a program that takes three digits from the user and adds the square of each digit.

16. Armstrong num

Write a program that checks whether a given number is an Armstrong number or not.

17. Narcissistic num

Write a program that will take user input of a (4-digit number) and check whether the number is a narcissist number or not.

18. Inhand salary

Write a program that calculates the in-hand salary after deducting HRA (10%), DA (5%), PF (3%), and tax. If the salary is between 5-10 lakh, apply a 10% tax; for salaries between 11-20 lakh, apply a 20% tax; for salaries above 20 lakh, apply a 30% tax. If the salary is between 0-1 lakh, print 'k'.

19. Menu driven

Write a menu-driven program with the following options: 1. cm to inch, 2. km to miles, 3. USD to INR, 4. exit.

20. Swap two nums

Write a program that swaps numbers.

21. Sum first n nums

Write a program to find the sum of the first n numbers, where n will be provided by the user. For example, if the user provides n = 10, the output should be 55.

22. Multiply without op

Write a program that can multiply two numbers provided by the user without using the * operator.

23. Factorial

Write a program that can calculate the factorial of a given number provided by the user.

24. First 25 odd nums

Write a program to print the first 25 odd numbers.

25. Prime number

Write a program to print whether a given number is a prime number or not.

26. Armstrong 100-1000

Print all the Armstrong numbers in the range of 100 to 1000.

27. Population calc

The current population of a town is 10,000. The population of the town is increasing at a rate of 10% per year. You have to write a program to find out the population at the end of each of the last 10 years. For example, if the current population is 10,000, the output should be like this: 10th year - 10,000 9th year - 9,000 8th year - 8,100, and so on.

28. Digit combos

Write a program to print all the unique combinations of 1, 2, 3, and 4.

29. Find hcf

The user will provide two numbers, and you have to find the HCF of those two numbers.

30. Calc lcm

The user will provide two numbers, and you have to find the LCM of those two numbers.

31. First 25 primes

Print the first 25 prime numbers.

32. First 20 fibs

Print the first 20 numbers of a Fibonacci series.

33. Compound interest

Write a program to calculate compound interest.

34. Sum n+nn+nnn

Write a program that accepts an integer (n) and computes the value of n + nn + nnn. n + nn + nnn ---> (2 + 22 + 222)

35. Count digits

Take a number from the user and determine the number of digits in it.

36. Find factors

Print all factors of a given number provided by the user.

37. Reverse num

Find the reverse of a number provided by the user (any number of digits).

38. Pyramid nested

Write a program to print the following pattern. * * * * * *

39. Patterns nested

To print different patterns using nested loops. * ** *** ****

40. Sum series

Write a program to calculate the sum of the following series up to the nth term: 1/1! + 2/2! + 3/3! + 4/4! + ... + n/n! The value of n will be provided by the user.

41. Series sum

Write a program to find the sum of the series up to the nth term: 1 + x^2/2 + x^3/3 + ... + x^n/n The value of n will be provided by the user.

42. Sum & avg

Write a program that continues to accept a number from the user until the user enters zero. Display the sum and average of all the numbers.

43. Simplify fraction

Write a program that accepts two numbers from the user, a numerator and a denominator, and then simplifies them. For example, if the numerator is 5 and the denominator is 15, the answer should be 1/3. For example, if the numerator is 6 and the denominator is 9, the answer should be 2/3.

44. String len without len()

Find the length of a given string without using the len() function.

45. Username from email

Extract the username from a given email. For example, if the email is saurabh@gmail.com, then the username should be saurabh.

46. Char occurrences

Count the frequency of a particular character in a provided string. For example, in the string 'hello, how are you,' the frequency of 'h' is 2.

47. Char index in string

Find the index position of a particular character in another string.

48. Count vowels

Count the number of vowels in a string provided by the user.

49. Remove char

Write a program that can remove a particular character from a string.

50. Palindrome string

Write a program that can check whether a given string is a palindrome or not.

51. Remove dupes list

Write a program to remove all duplicates from a list.

52. Title case without title()

Write a program to convert a string to title case without using the title() function.

53. Max value list without max()

Write a program to find the maximum item in a list without using the max function.

54. Reverse list

Write a program to reverse a list.

55. Search num list

Write a program to search for a given number in a list.

56. Square list items

Write a program that can create a new list from a given list, where each item in the new list is the square of the corresponding item in the old list.

57. Reverse words

Write a program that can reverse the words of a given string. For example, if the input is 'Hello how are you', the output should be 'you are how Hello'.

58. Word count string

Write a program that can count the number of words in a given string.

59. List sorted without sort()

Write a program to check whether a list is in ascending order or not.

60. Sep even-odd list

Create two lists from a given list where the first list will contain all the odd numbers from the original list, and the second one will contain all the even numbers.

61. Combine lists without '+'

Write a program to merge two lists without using the + operator.

62. Replace item

Write a program to replace an item with a different item if it is found in the list.

63. Flatten 2d list

Write a program that can convert a 2D list to a 1D list.

64. Union & intersection lists

Write a program that can perform union and intersection on two given lists.

65. Max item matrix row

The maximum item in each row of a matrix.

66. Matrix shape

Write a program to print the shape of a matrix.

67. Matrix mult possib

Write a program that can check whether you can perform matrix multiplication on two matrices.

68. Sort list without sort()

Write a program that can sort a given unsorted list. Don't use any built-in function for sorting.

69. Factorial without loop

Write a function that accepts a number and returns its factorial. You cannot use any loops.

70. Bag of words

Write a function that accepts a list of strings, performs Bag of Words, and converts it to numerical vectors.

71. Shortest dist coords

Write a program that accepts neighbors (a set of 2D coordinates) and a point (a single 2D coordinate) and tells the nearest neighbor (in terms of Euclidean distance).

72. Dict upper-lower chars

Write a function that accepts a string and returns the number of uppercase characters and lowercase characters as a dictionary.

73. List items to dict

Assume a list with numbers from 1 to 10 and then convert it into a dictionary where the keys would be the numbers of the list, and the values would be the squares of those numbers.

74. Most used word song

Write a program that can find the most used word in a song.

75. Merge dicts

Write a program to merge two given dictionaries.

76. Swap max min dict

Write a program to swap the key-value pairs for the maximum and minimum values. For example, if the dictionary is like this {‘a’: 1, ‘b’: 2, ‘c’: 3}: The output should be {a: 3, b: 2, c: 1}.

77. Login and reg

Write a dummy program that can perform login and registration using a menu-driven approach.

78. Int to string without str()

Write a program that can convert an integer to a string without using str().